A4.1 Transaction types and attributes

Slaves are classified as either:

Memory Slave

A memory slave is required to handle all transaction types correctly.

Peripheral Slave

A peripheral slave has an IMPLEMENTATION DEFINED method of access. Typically, this is defined in the component data sheet, that describes the transaction types that the slave handles correctly.

Any access to the peripheral slave that is not part of the IMPLEMENTATION DEFINED method of access must complete, in compliance with the protocol. However, once such an access has been made, there is no requirement that the peripheral slave continues to operate correctly. It is only required to continue to complete further transactions in a protocol compliant manner.

Note

- Compliant completion of all transaction types is required to prevent system deadlock, however, continued correct operation of the peripheral slave is not required.
- Because a peripheral slave is required to work correctly only for a defined method of access, it can have a significantly reduced set of interface signals.

The AXI protocol defines a set of transaction attributes that support memory and peripheral slaves. The **ARCACHE** and **AWCACHE** signals specify the transaction attributes. They control:

- how a transaction progresses through the system
- how any system-level caches handle the transaction.

In this specification, the term **AxCACHE** refers collectively to the **ARCACHE** and **AWCACHE** signals.

The following sections describe the transaction attributes:

- *AXI3 memory attribute signaling* on page A4-59
- *AXI4 changes to memory attribute signaling* on page A4-60.

Copyright © 2003, 2004, 2010, 2011 ARM. All rights reserved.
Non-Confidential